

Akhil Ram Shankar

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EDUCATION

University of California San Diego

San Diego, CA

B.S. in Computer Science and Mathematics

Sep. 2022 – Jun. 2026

- **Coursework:** OS, Database Systems, Computer Security, Machine & Deep Learning, Software Eng.
- **Awards:** Provost Honors, DiamondHacks 1st Place, SC23 MLPerf Winning Team ([HPC Wire](#) + [SDSC](#))

EXPERIENCE

Software Engineer

May 2024 – Present

UC San Diego – IT Services

San Diego, CA

- Authored 10+ stored procedures against million-row tables for a university-wide ERP migration, auditing and updating dependent procedures as schemas cascaded; mentored 4 student engineers on the methodology.
- First engineer at UCSD IT Services to deploy FaaS to production — built and owned end-to-end SDLC for two serverless workflow applications on AWS Lambda using API Gateway, CloudFormation, and CloudWatch, processing requests for 700+ student organizations.
- Built REST APIs and full-stack features across two enterprise web applications processing 7,000+ annual funding and storage requests; engineered Airflow ETL pipelines integrating 100,000+ row production datasets from multiple sources with 100% uptime during the ERP migration.
- Reduced data lookup latency by 81% (4 min → 45 sec) by replacing sequential record iteration with batch API calls, boosting productivity for 40+ administrative staff.

Systems Lead

Oct. 2023 – Present

RoboCup SSL, IEEE @ UC San Diego

San Diego, CA

- Architected a 6-layer real-time distributed control system for 6 autonomous robots spanning camera → Python AI → UDP/Protobuf → ESP32 → UART → STM32 → CAN bus, reducing latency from 200ms to 50ms.
- Led a 7-member subteam competing internationally in RoboCup SSL; Replaced bare-metal superloop with FreeRTOS and implemented omni-wheel inverse kinematics with per-motor PID tuning.
- Designed the full protocol stack — UDP multicast for stateless broadcast to N robots, Protocol Buffers for schema-defined serialization, and custom UART framing with header-detection for mid-stream resync.
- Built a bidirectional telemetry system streaming 8 real-time data streams per robot from STM32 → ESP32 → WebSocket → browser dashboard with live per-motor PID visualization.

PROJECTS

Cloud-Native Payments & Fraud Detection Platform | *Java, Rust, Python, Kafka, AWS, Kubernetes, Terraform*

- Built a distributed, event-driven payments platform using Kafka + SNS/SQS microservices on AWS EKS with idempotent APIs, autoscaling, and fault-tolerant architecture provisioned via Terraform pipelines.
- Implemented real-time fraud detection with PyTorch + FastAPI for sub-100 ms inference and integrated structured logging, distributed tracing, and metrics via Prometheus / Grafana for full observability.

Fault-Tolerant Distributed KV Store – Raft Protocol | *Rust, Tokio, Async I/O, Warp*

- Implemented a distributed key-value store in Rust using the Raft consensus protocol, with persistent logs, leader election, and crash recovery for linearizable reads/writes, tested under high-concurrency workloads.
- Simulated re-election and partition recovery across async multi-node clusters, achieving strong consistency and < 500ms failover under node crashes, with full state replication across all nodes.

CHIP-8 Emulator | *C++, SDL2*

- Implemented a low-level 8-bit emulator in C++ with full fetch-decode-execute cycle, opcode decoding, memory-mapped I/O, and stack-based control flow across 35+ opcodes.
- Emulated 60Hz CPU timing and frame-synchronized graphics rendering via SDL2, achieving cycle-accurate execution across 10+ classic ROMs.

TECHNICAL SKILLS

Languages: Java, Python, C, C++, Rust, TypeScript, C#

Systems & Tools: AWS (Lambda, S3, API Gateway, CloudFormation), Docker, CI/CD, SQL

Frameworks: Spring Boot, .NET, Node.js, Flask, FastAPI, Express, PyTorch, React

Concepts: Distributed Systems, Microservices, Multithreading, Networking